



CEB-MKIII Transparent batch reactor

The CEB-MkIII Transparent batch reactor is a double-skinned glass vessel with a one-litre internal working volume, fitted with a variable-speed agitator.

PRODUCT CODE: CEB-MKIII

Categories: CE Series - Chemical Engineering, CEB-MKIII Transparent Batch Reactor, Educational

Tags: Adiabatic , CE Series , CEB-MKIII Transparent Batch Reactor , Chemical Engineering Basic Process Principles , Chemical Reactors Teaching Equipment , Constant Temperature , Double-Skinned , Dyes , Exothermic Reaction , Isothermal , Saponification , Small Scale , Stirred Batch Reactor , Temperature Probes , Variablespeed Agitator

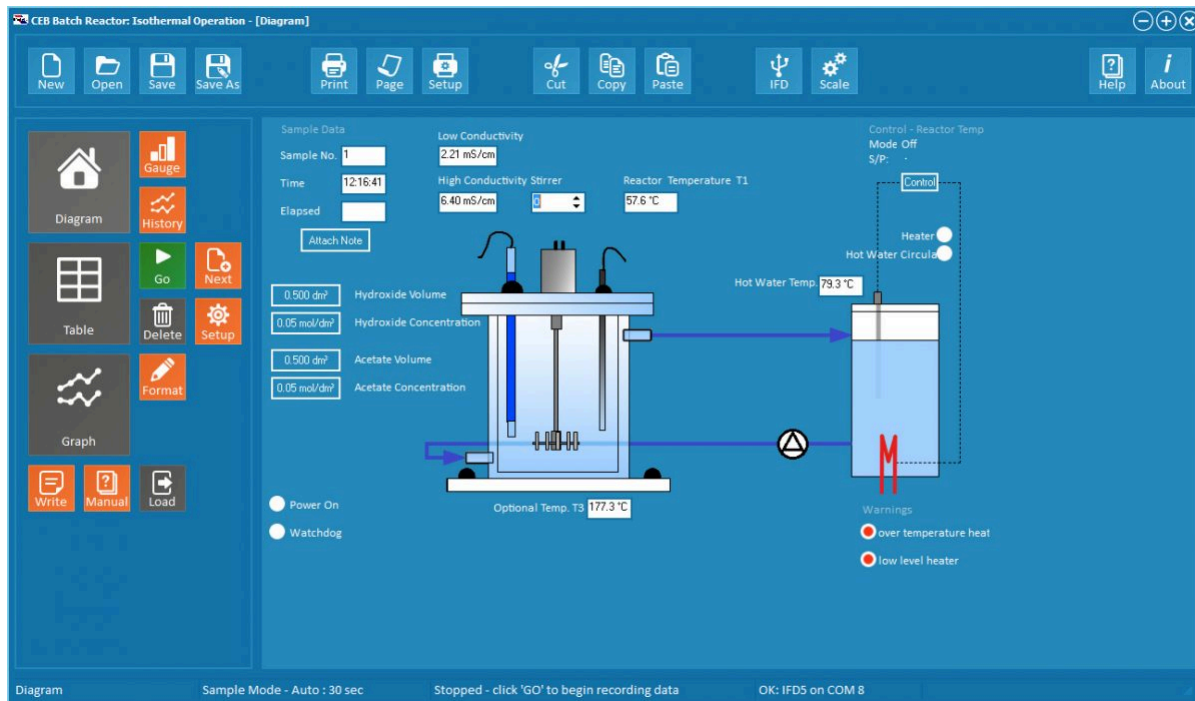
DESCRIPTION

The CEB-MkIII Transparent batch reactor is a double-skinned glass vessel with a one-litre internal working volume, fitted with a variable-speed agitator.

Hot water from the CEXC or cold water from the CW-17 can be circulated through the jacket for temperature control purposes maintaining the reactor contents at constant temperature.

Glands in the clear acrylic lid allow the CEXC conductivity and temperature probes to be fitted to facilitate monitoring of the reactions in progress such as the important saponification reaction. Isothermal and adiabatic operation reactions may be demonstrated. (Note, the isothermal reaction requires the Armfield CW-17 accessory if experiments at low temperature are to be studied or if the ambient temperature is high).

For adiabatic operation, the use of dyes enables the chemical reaction rates to be monitored visually by the change in colour at different degrees of conversion.



CEB-SoftwareScreen 600px

TECHNICAL SPECIFICATIONS

FEATURES & BENEFITS

Experimental content

- Isothermal Operation – To Determine the reaction rate constant in a stirred batch reactor
- Adiabatic Operation – To determine the rate equation for the hydrolysis of acetic anhydride to acetic acid in an adiabatic reactor.
- Investigation of the effect of reactant concentration on the reaction rate
- Investigation of the effect of temperature on conversion
- Visual monitoring of the chemical reactions
- Study of the temperature variation of an exothermic reaction on an adiabatic operation

Note: CEB-MkIII is designed for use with the CEXC service unit. CEB-MkIII can be used with the CEX service unit, but an additional STS5 temperature sensor will be required for complete functionality.

Ordering specification

- A small-scale batch reactor for use with the chemical reactors service unit designed to demonstrate both adiabatic and isothermal operation (CW-17 accessory is recommended for isothermal operation).
- 1L working volume
- The vessel includes a jacket through which hot water from the CEXC or chilled water from the CW-17 is passed. A variable-speed agitator aids heat transfer through the vessel
- The vessel is made of glass to give full visibility of the contents and enables the use of colour tracers to illustrate the reaction process
- Fitting points for temperature and conductivity sensors (supplied with the CEXC)

Requirements

Requires CEXC Base unit to operate

Consumables:

- 20g NaOH Sodium Hydroxide
- 50ml Ethyl Acetate
- 10mg Crystal Violet
- 100ml Acetic Anhydride
- 150ml Acetic Acid
- 25ml Sulphuric Acid

Shipping Specifications

PACKED AND CRATED SHIPPING SPECIFICATIONS

Volume: 0.1m³

Gross Weight: 5.6Kg

Dimensions

Length: 0.36m

Width: 0.33m

Height: 0.45m

Order Codes

CEB-MKIII

YOU MAY ALSO LIKE...



**CEY PLUG FLOW
REACTOR**

[Read more](#)



**CET-MKII
TUBULAR
REACTOR**

[Read more](#)



**CEZ LAMINAR
FLOW REACTOR**

[Read more](#)



**CEXC –
CHEMICAL
REACTORS
TEACHING
EQUIPMENT –
SERVICE UNIT**

[Read more](#)



**CEM-MKII
CONTINUOUS
STIRRED TANK
REACTOR
(CSTR)**

[Read more](#)



Armfield Limited. Unit 10 Headlands Business Park, Salisbury Road, Ringwood, Hampshire BH24 3PB.
England. Tel: +44(0) 1425 478781

